

RESEARCH ARTICLE

Shining a Light on Daytime Coral Spawning Synchrony Across Oceans

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ABSTRACT

Aim: The variety of coral taxa and environmental cues triggering broadcast spawning gave rise to contrasting theories about coral reproduction synchrony. Here, we shine a new light on this synchrony across oceans by highlighting how environmental cues modulate spawning time at various spatial scales in an understudied yet abundant gonochoric species.

Location: South Indian and Pacific Oceans.

Time Period: 2014–2023.

Major Taxa Studied: *Porites rus*.

Methods: *Porites rus* daytime spawning was investigated using a non-invasive citizen science approach (> 300 voluntary observers) at colony, reef, island and ocean scales. Spawning time was recorded between 2014 and 2023 at a total of 104 reef locations from 15 islands in three countries and multiple depths across the Indian and Pacific Oceans. Statistical models were employed to assess the relationships between spawning time and depth, light and sea surface temperature at different spatial scales, and in both male and female colonies.

Results: Spawning occurred synchronously for colonies located a few meters to > 15,000 km apart, monthly 5 days after full moon over an extended, uninterrupted period from October to April. Strong linear relationships between depth, light, water temperature and spawning time after sunrise held at the different spatial scales for both males and females, which spawned *ca.* 20 min apart. Interestingly, single colonies spawned across consecutive days and months.

Main Conclusions: The largest data set for a daytime coral species compiled here allows extremely accurate predictions of *P. rus* spawning months, days and time (minute-level precision) at different locations and depths in the Southern Hemisphere, facilitating field observations and experiments. Previously underexplored, the highly effective reproductive strategy of *P. rus* may explain its broad distribution and persistence in stressed environments, positioning it as an invaluable model organism for studying the physiological and genetic processes driving behavioural synchrony and biological rhythms across interconnected biogeographical regions.

1 | Introduction

Accurate synchronisation of conspecific behaviour benefits the maintenance of metapopulations through the completion of mutual goals, especially reproduction (Ims 1990). Across interconnected marine regions, sessile corals spawn synchronously to produce abundant planula larvae that disperse across reefs, maintaining sustainable metapopulation levels and genetic diversity worldwide. The reproductive timing of most corals is set at specific seasons and days of the year (Harrison et al. 1984; Babcock et al. 1986; Baird et al. 2009) when male, female or hermaphrodite broadcast spawning corals release gametes into the water column where mixing promotes fertilisation of eggs by sperm cells. This simultaneous coral gamete release on a given reef or series of reefs creates surface spawn slicks (bundles or eggs: Oliver and Willis 1987) or dense smoky clouds (sperm) that fog the sea for several minutes to hours and can even be observed from space (Yamano et al. 2020). While the production and release of these cells must synchronise among nearby colonies of the same species to ensure successful genetic mixing (Kitanobo et al. 2022), a species-specific timing (i.e., different coral species reproduce at various times of the year, lunar cycle and day or night times) also avoids coral species mixing and favours speciation (Fukami et al. 2003; Levitan et al. 2011; Baird et al. 2022).

Several environmental cues were identified to influence synchronous coral gamete maturation, gamete release synchrony and duration of the spawning season. Changes in sea surface temperature (Keith et al. 2016), solar and moonlight irradiance (Jokiel et al. 1985; Penland et al. 2004), periods of darkness (Lin et al. 2021; de la Torre Cerro et al. 2025), tidal cycles (Gouezo et al. 2020) or regional wind patterns (Sakai et al. 2020; van Woesik 2010) have been related to spawning processes. However, these interactions between environmental factors and spawning timing are often examined using geographically restricted, short-term data sets or investigated in isolation, limiting our ability to understand and generalise the multifactorial reproductive mechanisms across coral taxa. Recent findings also suggest that global change and anthropic activities may modify previous patterns in spawning synchrony, hence compromising coral reproduction (Shlesinger and Loya 2019; Davies et al. 2023). However, consistent long-term and large-scale data are still missing to accurately quantify these changes. Despite all the observations available to date (e.g., Coral Spawning Database, the first global spawning database initiative: Baird et al. 2021) and the reasonably accurate predictions available for several coral taxa (season and week-scale predictions), achieving high-precision forecasts (i.e., exact day and time) for individual spawning events remains challenging for most coral species (Baird et al. 2022; Komoto et al. 2023). Furthermore, the scarcity of data has given birth to a tangle of conflicting theories on the role of lunar cycle or temperature effects on spawning synchrony (Keith et al. 2016; Komoto et al. 2023; van Woesik et al. 2006).

The majority of coral reproduction studies focus on night-time coral spawning (Baird et al. 2022) while daytime spawning corals have received far less attention (Kinzie 1993; Plathong et al. 2006; Gouezo et al. 2020). Because night-time spawning is challenging to study due to observers' schedules and

logistics constraints, scientists have used proxies to assess spawning time, such as the collection of coral samples to examine gamete maturation under the microscope, and laboratory transplant experiments with artificial light and temperature regimes (Henley et al. 2022; Howells et al. 2014). These invasive alternatives to in situ observations are, however, particularly detrimental for small colonies and are both technically challenging and costly. Furthermore, if not performed at the precise timing, they can result in discrepancies in determining spawning times. In situ consistent long-term, real-time and species-specific observations at a variety of geographical scales to document spatially synchronised spawning are nearly nonexistent.

Taxon-wise, *Acropora* (Oken 1815) is by far the most intensively studied coral genus for spawning (Baird et al. 2022). Studies on other coral taxa are emerging but remain scarce and are even more data-limited (Gouezo et al. 2020; Baird et al. 2021). This includes *Porites rus* (Forskål 1775), for which infrequent and brief descriptive studies have been available only recently (Bronstein and Loya 2011; Gouezo et al. 2020). *Porites rus* is a gonochoric spawning coral species (Penland et al. 2004) found in the Indian and Pacific Oceans and in the Red Sea (Terraneo et al. 2021) from the near-surface to more than 65 m deep (Englebert et al. 2017). It colonises all habitats, including fringing reefs, channels and oceanic outer slopes, both in low and high current environments. It is abundant in many reefs of the Indian and Pacific Oceans, commonly forming massive coral heads in lagoons or large plates at greater depths, while branching and encrusting morphs also exist. This morphologic plasticity demonstrates its adaptation to different depths, current strengths and light intensities (Jaubert 1977; Forsman et al. 2009; Padilla-Gamiño et al. 2012). Finally, it is highly resistant to thermal (Lenz and Edmunds 2017), pH (Lenz and Edmunds 2017), sedimentation (Padilla-Gamiño et al. 2012; Golbuu et al. 2008) and light (Hoadley et al. 2021) stresses. Corals presenting these properties have commonly been considered as 'winners' in front of global change, with the potential to outcompete and replace other coral taxa in the future (Loya et al. 2001; van Woesik et al. 2011; McWilliam et al. 2020). Nevertheless, this potential has not been comforted for the majority of these 'winning' species by a better knowledge of their spawning spatiotemporal patterns and processes.

To address this knowledge gap, we analysed data on *P. rus* spawning occurrences collected since 2014. These data were obtained through citizen science, which—when based on clear and simple protocols—is a cost-effective yet powerful means for gathering large volumes of reliable observations to support scientific research (de Sherbinin et al. 2021). Observations of successful spawning events during the initial years (2014–2019) led to the creation, in November 2021, of a non-profit organisation and an associated monitoring network. This initiative aimed to systematise the long-term collection of *P. rus* spawning data by coordinating coral spawning monitoring with citizen participation, sharing results and fostering public interest. As a result, the data spanning 2014 to 2023 were compiled with the involvement of several hundred citizen scientists of all ages across three jurisdictions and two oceans. To date, this effort has produced the most extensive data set ever assembled on *P. rus* spawning.

We used this dataset to explore the spatial scale at which spawning synchrony occurs and identify the possible environmental triggers. First, given that *P. rus* is a daytime-spawning coral, we investigated if light plays a key role in determining the timing of gamete release, and if light attenuation can account for variations in spawning time across different depths. Secondly, we examined whether temperature, which increases along the spawning season and may vary between sites, can modulate spawning time at different spatial scales and for both sexes. These insights can contribute to enhance the accuracy of spawning time prediction based on the geographic and environmental characteristics of a site at a broad scale, using data from two oceans.

2 | Methods

2.1 | Reef Scale Spawning Observations

Opportunistic observations of *P. rus* spawning on 11th November 2014, 8th November 2017, and 27th November 2018 in Moorea and Tahiti (French Polynesia) and the only available publication detailing *P. rus* spawning for the Southern Hemisphere (Bronstein and Loya 2011) allowed us to identify the possible spawning day and time. Accordingly, on 17th November 2019, four sites on two French Polynesia islands were synchronously surveyed, collecting data that confirmed the predicted and reported (Bronstein and Loya 2011) spawning time accuracy and synchrony.

On this basis, the organisation developed an in situ, noninvasive observation protocol allowing anyone to observe *Porites rus* spawning at any location in the Southern Hemisphere where *P. rus* is present (Figure 1a). Observation sites comprising *P. rus* colonies should be around 50 × 50 m large to ensure that all colonies are visible and can be reached rapidly when swimming from one end to the other. Site GPS coordinates or exact location on a map must be recorded. At a specific day of the month and time after sunrise calculated by the organisation, the observer enters into the water 10–15 min before the estimated spawning time, wearing a waterproof watch set up to the exact local time and recording the observation date (yyyy-mm-dd). The exact time (hh.mm) at which the first colony on the site is observed to spawn is recorded. If spawning has already started on the site when the observer enters the water, or if the observer is not confident in his spawning observation exact time, the spawning time recorded is categorised as ‘approximate’. The time when the last colony of the site finishes spawning is recorded and categorised as ‘exact’ or ‘approximate’. In case no colony is observed to spawn on the whole site, the absence of spawning is recorded.

Following this protocol, data were gathered between 2014 and 2023 across three jurisdictions of the Southern Hemisphere (Figure 1a) by > 300 volunteer observers (citizen scientists and, occasionally, scientists) and transferred to the organisation by e-mail or directly on a dedicated mobile application. The organisation proceeded to a data check-up before validating each observation for integration into a reef-scale (or site) data set compiled for the purpose of this study. Initial advice on data gathering were kindly provided by the Coral Spawning Database managers (Baird et al. 2021). Observations from 2014 to 2019 were

performed between 0 and 5 m depth, and, because *P. rus* also colonises greater depths (Englebert et al. 2017), from 2020 the monitoring was extended to deeper reefs (6–30 m) using SCUBA.

2.2 | Individual Colony Spawning Observation

To better understand spawning time variability at the individual level, 115 *P. rus* colonies were tagged and monthly monitored by volunteer observers from 2021 to 2023 (two spawning seasons: 2021–2022 and 2022–2023) in selected sites at 1–2 m depth: 30 colonies in Vairao, 25 in Venus Point and 30 in Arue (Tahiti Island), 10 in Maatea (Moorea Island), and 20 in the Educative Marine Area (Bora Bora Island). Colony monitoring was similar to site monitoring described above, but spawning start and end times were recorded for each tagged colony separately. Since gonochoric coral colonies may sometimes exhibit hermaphroditic behaviour (Neely et al. 2018), sex was individually verified at each spawning event by trained volunteers who had received specific instruction. These data were gathered in a colony-scale dataset.

2.3 | Photosynthetic Active Radiation (PAR) Recording

Light effect has been evidenced for other coral species (daytime or night-time spawners) and complex theories involving diurnal light cycles, period of darkness after sunset, lunar photoperiod, nocturnal illumination or twilight colours inducing gamete release were proposed, without any consensus reached to date (Mendes and Woodley 2002; Babcock et al. 1986; Sweeney et al. 2011; Boch et al. 2011; de la Torre Cerro et al. 2025). Because *P. rus* lives at different depths characterised by different light regimes, we investigated whether *P. rus* needed to receive a certain quantity of light to induce gamete release (Vize et al. 2008; Brady et al. 2009; Bouwmeester et al. 2023), as shown with moonlight for night-time spawning species (Jokiel et al. 1985; Boch et al. 2011; Rosenberg et al. 2017).

To test this, two Odyssey photosynthetic irradiance (Photosynthetic Active Radiation (PAR): 400–700 nm) loggers were installed on the 12th November and 26th November 2022 at, respectively, 2 m and 21 m depth near *P. rus* colonies at the Vairao monitoring site, in an unshaded area (Figure S1). They were calibrated to record PAR irradiance at 10-minute intervals during the spawning season. Loggers were removed on the 4th March and 10th April 2023, respectively. For each spawning date, we cumulated the irradiance values monitored every 10 min between sunrise (i.e., just after dark when irradiance becomes > 0) and spawning time for the shallow and deep colonies, then averaged the spawning start time and the cumulated irradiance per date.

2.4 | Sea Surface Temperature Data and Monitoring

Because the spawning season takes place during the Southern Hemisphere warm season (tropical spring and summer), we hypothesised that the seasonal increase in sea water

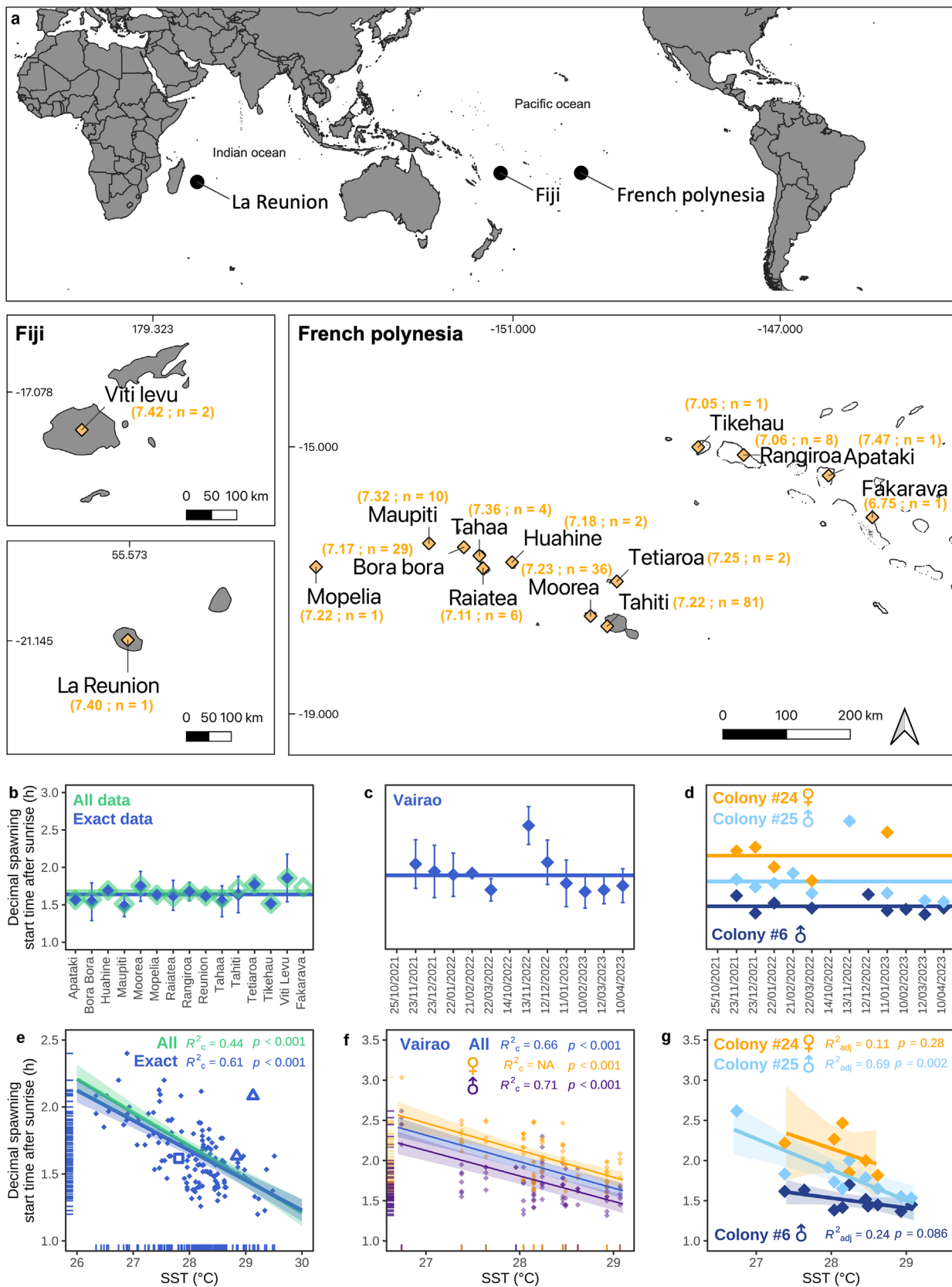


FIGURE 1 | Legend on next page.

temperature may modulate the exact spawning time after sunrise. Previously, the literature reported the increase in sea surface temperature (SST) as a trigger for the onset of the

spawning season (Keith et al. 2016; Gouezo et al. 2020; Osman et al. 2024). Here, we investigated the influence of temperature at a daily scale (i.e., considering temperature on the exact

FIGURE 1 | *Porites rus* spawning is synchronous and modulated by temperature. (a) *P. rus* decimal spawning start time (local time in decimal hours) at each island, averaged across shallow sites and spawning occurrences ($n = 5$ DAFM between 2014 and 2023, based on the time zone corresponding to each location). (b) Mean decimal spawning start time after sunrise (in decimal hours) for all monitored islands between 2014 and 2023 ('La Reunion': La Reunion Island in the Indian Ocean; 'Viti Levu' in Fiji; others: French Polynesian islands) at shallow sites using exact data (blue symbols; SD represented as vertical bars) and all data (exact and approximate data: green symbols; SD not shown for clarity). The horizontal lines represent the decimal spawning start time after sunrise averaged across all corresponding observations (exact and all data). (c) Mean and SD of exact decimal spawning start time after sunrise for shallow colonies ($n = 30$) monitored at Vairao, Tahiti. The horizontal line represents the decimal spawning start time after sunrise averaged across all observations. Note that no spawning occurred in October 2021 and 2022. (d) Exact decimal spawning start time after sunrise for one female and two male colonies monitored in Vairao, Tahiti. No spawning occurred where symbols are absent. The horizontal lines represent the decimal spawning start time after sunrise averaged across the colony observations; note that these two example colonies spawned on average earlier than the Vairao average (c). Linear mixed-effect models (LMM) of the relationship between sea surface temperature (SST in °C) and spawning start time after sunrise at large (e: blue symbols and fit: exact data; green fit: exact and approximated data; triangles: Fiji ($n=2$); square: La Reunion ($n=1$) data), site (f: orange symbols and fit: females; dark blue symbols and fit: males; blue fit: all data) and colony (g) scales. LMM were fitted to the data with sites (e) and colonies (f) as a random factor. Linear models were fitted to the data in (g). Shadings represent the LMM and LM 95% confidence intervals.

day of spawning) on the exact spawning time after sunrise at the minute-level resolution throughout the reproductive season as temperature gradually increased. Our goal was to better understand the variability in spawning time across months to enhance minute-scale spawning time predictions throughout the season.

Daily foundation SST (i.e., free of diurnal variability) at high resolution ($0.05^\circ \times 0.05^\circ$) for all sites and dates recorded in the reef-scale and colony-scale data sets was collected from the OSTIA (Operational Sea Surface Temperature and Ice Analysis: Good et al. 2020) global sea surface temperature reprocessed product database on Marine Copernicus (<https://marine.copernicus.eu>). To validate the use of these oceanic SST data for lagoon sites, lagoon in situ temperature data were collected from loggers (multiparameter probe NKE model Smatch and Sambat and temperature probe Seabird SBE 56, installed successively at the same location at 5 m depth) located near the Vairao monitoring site: OSTIA SST and in situ temperature data showed a significant correlation ($RMSE = 0.21$, $R^2 = 0.95$, $p < 0.001$; see Figure S2 in Supporting Information), showing that oceanic OSTIA SST were reliable values to use in our analysis. OSTIA SST were therefore subsequently assigned to each spawning date and location in the reef-scale and colony-scale spawning data sets to assess the effect of spawning day SST on start time after sunrise throughout the spawning season, at different spatial scales and for both colony sexes.

2.5 | Statistical Data Analysis

Because citizen science data may carry potential biases in accuracy (Aceves-Bueno et al. 2017; Jamodiong et al. 2018), we considered two sets of data: the 'exact' data set included only observations validated as accurate, while the 'all' data set included both exact and approximative observations. This distinction was applied in the site-scale analyses described below, when relevant. To evaluate whether spawning start time after sunrise varied between these two datasets, a Student *t*-test was performed.

To test whether spawning time remained consistent across spawning seasons, a one-way analysis of variance (ANOVA) was conducted on the mean monthly exact spawning time after

sunrise using 'season' as a categorical variable with three levels ('2020–2021', '2021–2022' and '2022–2023').

To demonstrate the consistency of spawning time relative to sunrise and to highlight the time lag in spawning start time between the shallow lagoon and the deep channel-outer slope habitats, we fit a linear model (LM) to the exact spawning time for each habitat as a function of time of sunrise. Then, to estimate how the spawning start time after sunrise changes with depth, we fit two LMs (one for 'exact' only data, and one for 'all' data) between depth and spawning start time after sunrise. The adjusted *R*-squared and the *p* values were calculated.

To study the relationships between temperature and spawning start time after sunrise at different spatial scales (Southern Hemisphere and reef scale), linear mixed-effects models (LMM) were implemented using SST as a fixed effect and site or colony as a random effect. To assess this relationship at the colony scale, LMs (no random effect) were used. Corresponding conditional and adjusted *R*-squared values and *p* values were calculated.

Finally, we tested for the difference in male and female spawning start time after sunrise using a Student *t*-test.

All analyses were run using R 4.4.3 (R Core Team 2025) and the packages 'raster' (Hijmans 2022), 'lme4' (Bates et al. 2015) and 'lmerTest' (Kuznetsova et al. 2017). Maps were constructed on QGIS 3.16.

3 | Results

3.1 | Citizen Science Reveals *Porites rus* Spawning Schedule and Large-Scale Spatial Synchrony

Porites rus appeared to follow a highly synchronised spawning pattern across different scales (Figure 1), spawning 5 days after full moon (DAFM), ca. 1.5 to 2 h after sunrise in fringing reefs (i.e., 1–5 m depth) (Figure 1a). Total lagoon observations collected between 2014 and 2023 using the citizen science approach described in the Methods revealed that spawning day and start time (local time and start time relative to sunrise) were extremely synchronised over 15,000 km (Figure 1a), at 104

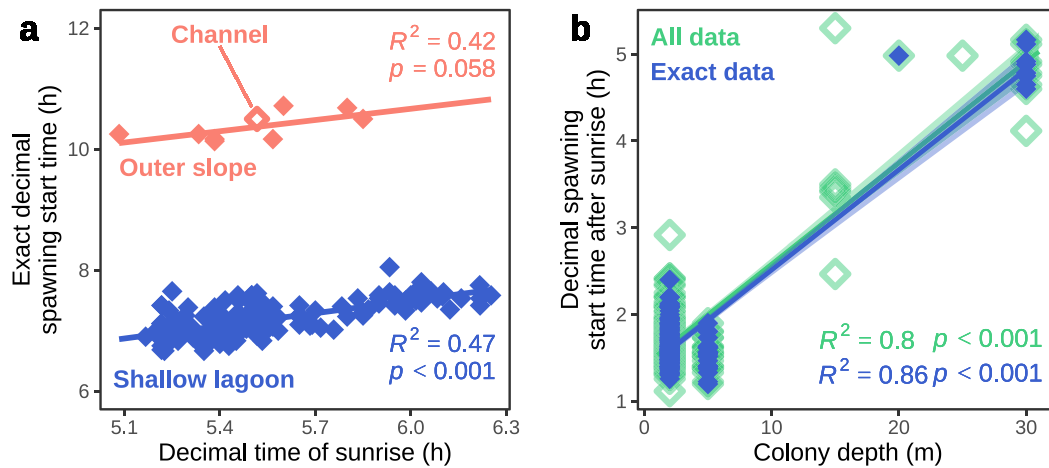


FIGURE 2 | Spawning time varies with depth. Linear models of the relationship between (a) decimal time (in hours) of sunrise and exact decimal spawning time per habitat ('Outer slope' and 'Channel' data, located at similar depths, are combined in the same model) and (b) reef location 5-m-depth bins and spawning start time after sunrise using exact (blue symbols) and all (exact and approximate: green symbols) data. Shadings represent the 95% confidence intervals of the linear models.

reefs spread over 15 islands (Figure 1a,b: from Reunion Island in the Indian Ocean to Fakarava Atoll in the Pacific Ocean: $\text{mean}_{\text{exact}} = 1.64$, $\text{min}_{\text{exact}} = 1.20$, $\text{max}_{\text{exact}} = 2.40$, $\text{SD}_{\text{exact}} = 0.23$ decimal hours after sunrise), as well as for all colonies tagged at the monitoring sites (e.g., Vairao, Tahiti: Figure 1c: $\text{mean} = 1.89$, $\text{min} = 1.32$, $\text{max} = 3.03$, $\text{SD} = 0.33$ decimal hours after sunrise; other sites: see Figure S3a–d).

To explore the duration of the phenomenon within a year, sites and colonies monitored across 9 months (exception: July, August and beginning of September) from 2020 to 2023 evidenced that *P. rus* spawning season extends from October at the earliest to April at the latest, at the same time of the 5th DAFM each month (Figure 1c,d and Figure S3). No spawning was detected in May, June and late September at any of the monitored sites, and the season was shorter locally (e.g., no spawning in October in Vairao, Tahiti Island: Figure 1c). Individual colonies spawning time was highly consistent across months and seasons (Figure 1c and examples of three tagged colonies in Vairao, Tahiti: Figure 1d; other sites: see Figure S3e–h). Interestingly, we did not detect any change in mean spawning time between the monitored seasons (one-way ANOVA: $F(2) = 2.094$, $p = 0.154$). Analyses of the two data sets therefore showed a strong consistency in spawning time after sunrise 1) at transoceanic, reef and colony spatial scales, 2) during the whole spawning season and 3) across seasons (Figure 1b–d).

Monitoring specific colonies in Tahiti, Moorea and Bora Bora allowed characterising gamete release duration precisely, a phenomenon lasting 30.6 min on average ($\text{SD} \pm 13.2$ min, males and females combined). Additionally, although we focused our observations on the 5th DAFM, we observed that several colonies spawned on the 6th DAFM during the 2022–2023 season at the same scheduled time. Some of these colonies were identified to spawn both on the 5th and on the 6th DAFM, while others spawned only on the 6th DAFM.

Despite a variability in the precision of observed spawning time, the use of both approximate and exact data in the

analyses did not result in significant differences in mean spawning time between the 'exact' and 'all' data groups (Student *t*-test: $t(332.74) = -1.5884$, $p = 0.1131$; Figure 1b).

3.2 | Depth and Light Effects on Spawning Time

When assessing *P. rus* spawning time at greater depths, unsuccessful attempts through trial and error in Tahiti, Moorea, and Takapoto Atoll (French Polynesia) between 7 and 9 a.m. were made, until a coincidental observation on 23rd November 2021 on Tikehau Atoll (French Polynesia) forereef at 15 m depth confirmed spawning at ca. 10 a.m. local time. Subsequent attempts to observe spawning at deep sites were successfully scheduled around 10 a.m. in deep lagoons, passes, and forereefs.

Colonies in deeper habitats (lagoon channels and forereefs) spawned on average 3.25 h after shallow colonies (Figure 2a). Deviations from the linear model did exist, but we did not detect outliers in any of the habitats (Figure 2a). Similar linear relationships with depth were found when using either the 'exact' or the 'all' data groups ($R^2 = 0.88$, $p < 0.001$ and $R^2 = 0.82$, $p < 0.001$, respectively: Figure 2b). These results indicate that *P. rus* was extremely well synchronised with sunrise depending on its depth. These highly significant linear relationships can be used to predict the spawning start time after sunrise accurately for sites located at different depths (see Appendix S4), although more work is required to better characterise the spawning time variability observed around 15 m depth (Figure 2b).

Since the intensity of light decreases with water depth, we hypothesised that the delay in the onset of spawning at greater depths (Figure 2a,b) resulted from the diminished intensity of penetrating light, causing the gamete release to be postponed until the colonies had accumulated an adequate amount of light. Accordingly, we observed that: (1) mean spawning time was on average earlier when cumulated irradiance was high (e.g., January and February 2023); (2) the first spawning time of the season (November 2022) occurred later than the others,

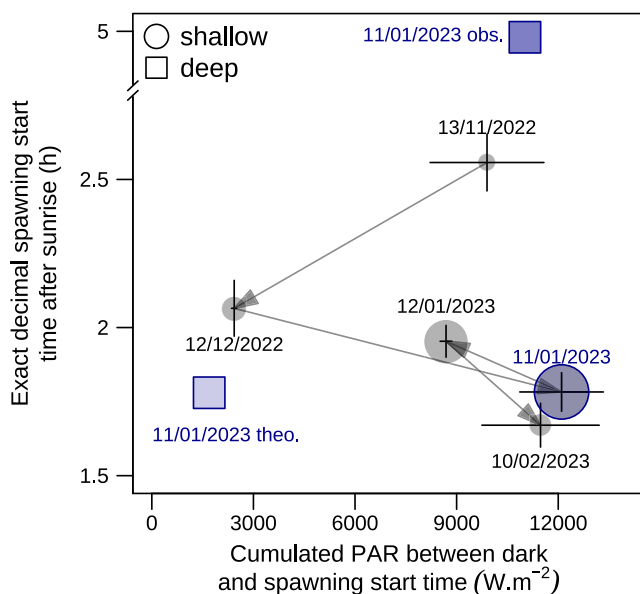


FIGURE 3 | Cumulated photosynthetic active radiation (PAR) irradiance (W.m^{-2}) and exact spawning time after sunrise (in hours) for *Porites rus* colonies in Vairao (Tahiti Island) during the 2022–2023 spawning season at low and high depths. Arrows indicate the temporal trajectory for shallow colony spawning, and the spawning date is indicated near each symbol; black bars in each symbol indicate horizontal and vertical SE. Spawning on 11/01/2023 for shallow and deep colonies is highlighted in blue; 'obs': spawning observed for the deep colony; 'theo': theoretical cumulated PAR irradiance if the deep colony had to spawn at the same time as the shallow colonies on this day.

requiring higher irradiance (a delay that is systematically observed every year at each site, see Figure 1c); and (3) spawning at the deep site started only when the colonies received nearly as much cumulated irradiance at that depth as the shallow colonies, that is, 5 h against 1.78 h after sunrise, with 11,017 cumulated W.m^{-2} against 12,098 cumulated W.m^{-2} , respectively (Figure 3).

Finally, we calculated the cumulated irradiance for the deep colonies if they had to spawn at the same time as shallow colonies, that is, 1.78 h after sunrise. This theoretical irradiance was under the mean lower limit observed during the season (on 12/12/2022), and nearly 10 times lower than the irradiance received at that date by shallow colonies (on 11/01/2023) (Figure 3).

3.3 | Temperature and Spawning Time Relationship Across Scales

A strong significant linear relationship revealed that spawning start time after sunrise occurred earlier at higher SST across oceans (LMM on exact data: $p < 0.001$, $R^2_{\text{conditional}} = 0.61$, Figure 1e), for colonies within a site (LMM on Vairao colonies: $p < 0.001$, $R^2_{\text{conditional}} = 0.66$, Figure 1f) and for individual colonies (LM on individual colonies, Figure 1g). This indicates that *P. rus* spawned, on average, earlier after sunrise at warm locations, and that spawning occurred earlier at a given location and for single colonies when the ocean warmed up along the spawning season (tropical spring and summer) (Figure 1c,d).

3.4 | Synchrony Between Males and Females

Gonochoric *P. rus* male spawning resembles a white opaque cloud of smoke (Figure 4c) while female spawning is similar to a transparent cloud of < 1 mm round particles going toward the surface (Figure 4d). Sites naturally varied in their proportion of male and female colonies with an average of $68\% \pm 20\%$ males and $32\% \pm 20\%$ females per site among the tagged colonies. In Vairao, the proportion of colonies that spawned was well-balanced at most months (Figure 4a) and females started to spawn 19.8 min after males (1.72 ± 0.28 h after sunrise for males, 2.05 ± 0.28 h after sunrise for females; Student t -test ($t(148.67) = -7.11$, $p < 0.001$, Figure 4b; other sites: see Figure S5). Data distribution also indicated that female spawning started during the male spawning peak, then spawning reduced rapidly for males and progressively for females (Figure 4b). Additionally, both male and female spawning start times were strongly and similarly correlated with SST, from the reef scale (Figure 1f) down to the individual scale (Figure 1g).

4 | Discussion

We showed that *P. rus* spawning occurs monthly from October to April in the Southern Hemisphere at predefined days and times, an unprecedented spawning season duration for a coral (Baird et al. 2022). While coral spawning was known to vary within a range of days (e.g., between the 1st and 11th DAFM: Levitan et al. 2011; Lin and Nozawa 2017) and some coral species were observed to spawn over consecutive months or days at the reef level (Lin and Nozawa 2017; Jamodiong et al. 2018), the literature remains evasive on the capacity of a single colony to spawn over consecutive months and days. Our colony-scale data analysis revealed that *P. rus* single colonies can spawn for up to five consecutive months, which has not yet been reported to date for an entire single colony, neither for *P. rus* nor for other coral species (Baird et al. 2022; Gouezo et al. 2020). Additionally, we showed that *P. rus* colonies have the potential (i.e., the gamete storage capacity and the energy to produce and expulse them) to spawn 2 days in a row. This consecutive spawning should be the focus of further work to assess whether some colonies fail to capture the lunar cycle cue and spawn only after the 5th DAFM, or if all individuals have the capacity to spawn several days in a row. The monthly and daily recurrence of colony spawning within a single season highlights *P. rus* strong reproduction efficiency, contrary to other corals (Baird et al. 2022), and could explain its large-scale distribution and persistence in stressed environments. Finally, we accurately quantified the delay between male and female spawning, which has been evidenced for other gonochoric coral species and is supposed to increase fertilisation success rate (Vize et al. 2005; Neely et al. 2020).

Animal reproduction is known to be largely influenced by light and temperature (Beltran-Frutos et al. 2022; Dougherty et al. 2024), two factors responsible for gamete maturation and expulsion in corals (Nozawa 2012; Crowder et al. 2014; Wuitchik et al. 2019; Komoto et al. 2023). Here, we showed that the increase in temperature during the spawning season (tropical spring and summer) induced earlier spawning at large scale as well as for single colonies, highlighting that temperature influences coral physiology throughout the season,

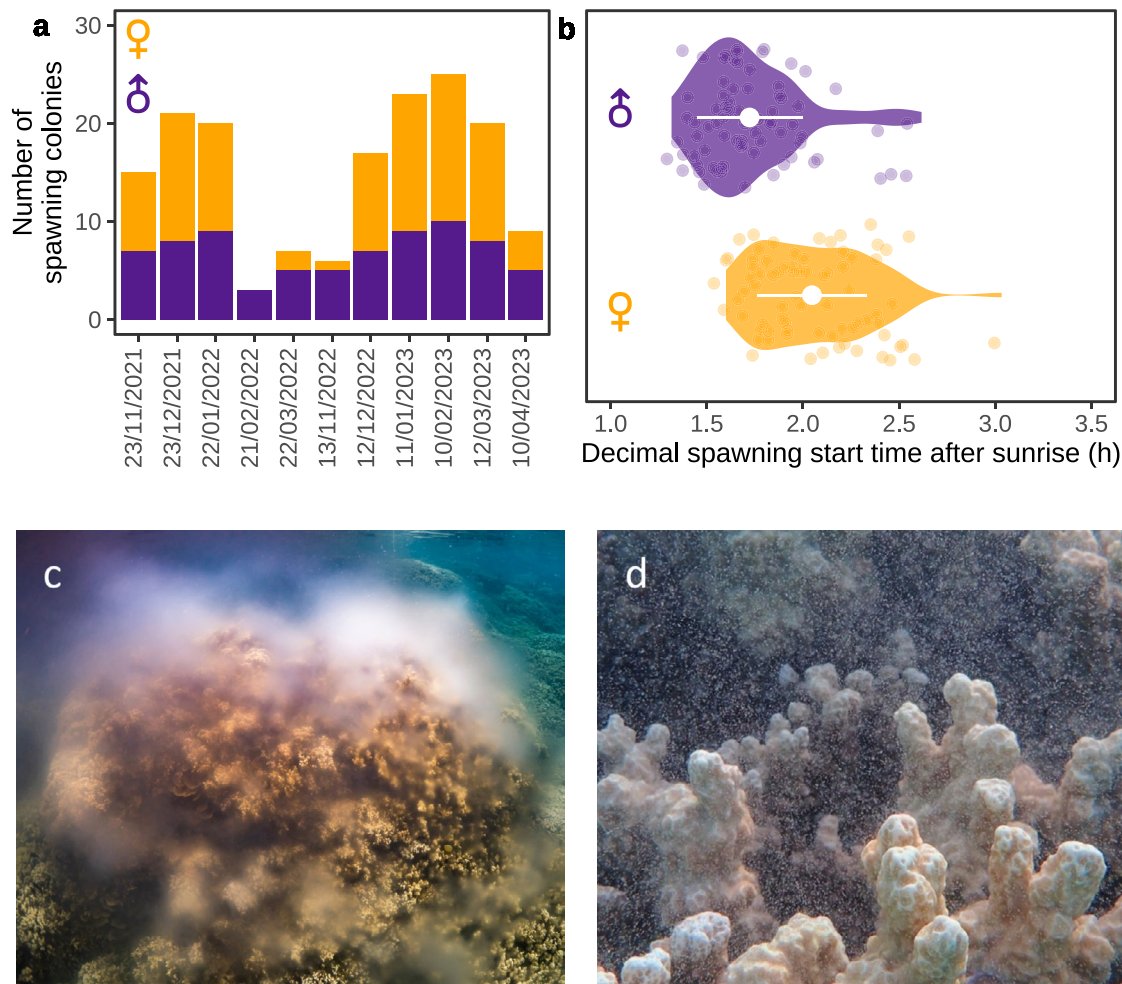


FIGURE 4 | Spawning observations and time difference between *Porites rus* male and female colonies. (a) Number of tagged male and female colonies spawning in Vairao, Tahiti Island, between 2021 and 2023. (b) Decimal spawning start time after sunrise (in hours) at Vairao site for male and female colonies. White dots indicate the mean start time after sunrise and horizontal white lines show SD. Females at Vairao spawn on average 19.8 min after males. Male (c) and female (d) spawning.

probably resulting from the increase in the speed of chemical reactions in reproductive organs and cells at high temperatures (Crowder et al. 2014) or from a thermally variable regulation of the expression of genes associated with reproduction (Wuitchik et al. 2019). A relationship with temperature has been evidenced at two sites in Taiwan (Lin and Nozawa 2023), but the present results also hold for single colonies as much as for sites situated 15,000 km apart. Furthermore, comparable temperature-spawning time relationships for males and females suggest that both sexes undergo similar temperature regulation processes throughout the season. Given the geographic imbalance in the dataset (e.g., one data point from La Reunion, Indian Ocean), additional data collection is warranted to further support and strengthen these large-scale findings. To further investigate this relationship on a 3D level, deploying temperature loggers down to the deepest locations of *P. rus* (i.e., >65 m, Englebert et al. 2017), where temperature is likely to be lower than at the surface (Thomas et al. 2010; Van Wynsberge et al. 2020), would allow to assess whether temperature vertically influences spawning time. Light, on the other hand, acts as a trigger for gamete expulsion. However, this has mainly been highlighted for night-time spawning coral species with moonlight (Komoto

et al. 2023). Despite our PAR logger data were limited to one season, the daylight-spawning time relationship found here is encouraging to extend monitoring of individual colonies at intermediate and high depths to gather additional data to support the cumulated PAR irradiance hypothesis for daytime spawning corals. *In fine*, these additional temperature and light data could enable the development of a predictive, multifactorial model that integrates both variables to accurately estimate spawning time across different geographic locations (horizontal scale) and depths (vertical scale).

Daytime spawning corals are far less studied than night-time spawning corals. To assess the possible role of sunset commonly referenced in coral spawning studies (Lin et al. 2021), we examined the sunset time (i.e., on the evening preceding the spawning morning) using our site-scale data set. While the relationship between sunset and spawning time was statistically significant ($p < 0.001$), the associated R^2 value was very low (0.08). Similarly, we found no relationship between moonrise time and spawning time. These results were therefore not presented in this analysis. In contrast, the relationship between sunrise and spawning time was substantially stronger ($R^2 = 0.39$). Although this remains a

correlation and does not confirm a causal mechanism, sunrise therefore appears to be a robust predictor of spawning onset and is currently the most promising candidate for identifying the environmental cue that triggers gamete release for *P. rus*.

The overall spawning phenomenon of *P. rus* occurring over thousands of kilometres is highly synchronous and follows definite cues that we have partially identified, but interindividual (i.e., among colonies) and intersite variations in spawning time and date (e.g., 6th DAFM) do exist. While this natural intraspecific variability may constitute a means of increasing spawning duration at the reef scale to enhance fertilisation, genetic mixing and genetic variability (Levitán et al. 2011; Oomen and Hutchings 2015), the broader variation in spawning times and dates observed at other locations (e.g., Penland et al. 2004: 4th DAFM 9:25 pm; Bronstein and Loya 2011: 5th DAFM 11:00 am) raises the possibility that *P. rus* may represent a species complex, with cryptic species exhibiting similar morphologies. This question could be addressed by conducting genetic analyses of *P. rus* samples collected across the ocean basins where the species occurs, in both the Northern and Southern Hemispheres. Moreover, the physiological and genetic processes (Celis et al. 2016) inducing gamete maturation and synchronised release should be investigated to assess whether spawning is induced individually by environmental cues, genetic precision and/or conspecifics molecular signalling within a site. This could provide additional insights into whether certain colonies in specific locations exhibit a regulated delay in spawning or, alternatively, a dysfunction in their spawning physiology due to genetics. Such genetic-focused research should be conducted in conjunction with geographic (e.g., depth, latitude and proximity with the equator) and environmental (e.g., light, temperature, sun and moon, turbidity, salinity, tide, wind) factors, which vary considerably across the existing studies (this study; Bronstein and Loya 2011; Penland et al. 2004; Gouezo et al. 2020).

This work provides evidence that well-organised citizen science can contribute to accumulating valuable and accurate data to study large-scale biological phenomena (de Sherbinin et al. 2021). Similar patterns were found for exact-only and all spawning data, supporting the validity of using unexperienced volunteers to collect reliable scientific observations. Monitoring sites in many Indo-Pacific islands are effortlessly accessible, and the spawning window at the beginning of the day can meet anyone's agenda (see Appendix S4). A bias regarding the exact time of spawning does exist at intermediate depths (ca. 15 m), but we showed here that it can be largely thwarted by the accessibility of the extremely simple protocol (see Section 2 and Appendix S4) which attracts a large number of observers (who are offered, due to the high predictability of the spawning time, a rewarding experience in the field compensating for their voluntary work) and thus data, at large-scale and at no cost. As an example, the largest synchronous spawning in the database was observed on 23rd December 2021 by 115 volunteers recording 30 simultaneous spawning incidences at 32 reefs over 9 islands spread over 360 km. Gathering the same amount of data with classical scientific means would have been impossible without consistent funding.

Predicting coral spawning is crucial to limit the impacts of coastal development on coral reefs. Understanding the

spawning processes and triggers of a widespread species can precisely inform decision making and spatial management at local to large scales. Despite a few countries (e.g., Australia: Styan and Rosser 2012) have followed well-established precautionary rules for decades, they still face a lack of accuracy in predicting the exact day of coral spawning. In the case of *P. rus*, which consists of a large proportion of the fringing coral community in many tropical countries, spawning is sufficiently predictable for clear precautionary guidelines to be set up and followed. Locally, the organisation is actively encouraging coastal and marine construction works not to alter the marine environment (e.g., no digging, no mining, no river curcation, no turbulences, no underwater noise disturbance) during the spawning days across the entire, highly predictable spawning season. Alerting about the fragility of coral reefs, especially in the fringing reef area, is crucial and has proven to be efficient, as local construction companies are now following guidelines and asking for environmental impact advice before implementing their works. Raising awareness about environmental fragility by engaging the public through field observations can be an effective and direct way to protect the environment before new, appropriate laws are enacted and enforced by governments.

5 | Conclusion

In the context of global coral reef degradation, our findings bring significant input to foster coral biological clock research. Global change is suspected to alter synchronic ecological processes at a large scale (Hackett-Pain and Bogdziewicz 2021; Karatayev et al. 2025), including coral spawning synchrony (Shlesinger and Loya 2019). So far, we have observed no change in *P. rus* spawning time over the years. This constancy in spawning synchrony may be due to *P. rus* high resistance to environmental and anthropic stresses (Golbuu et al. 2008; Padilla-Gamiño et al. 2012; Lenz and Edmunds 2017; Hoadley et al. 2021), and accumulating data over additional years will help determine whether the synchrony of *P. rus* gamete release remains stable in the future. The absence of detecting long-term changes could mean that *P. rus* is insensitive to environmental changes or able to adapt its reproductive processes to these changes, hence procuring an additional characteristic of a 'winner' species in front of global change. We therefore believe that more attention should be given to *P. rus*, as it is an accessible coral that has long been overlooked. Despite the study of *P. rus* spawning being only in its early stages here, our results on the spawning time relationship with depth, light, and temperature at various spatial and time scales can catalyse research for *P. rus* around the globe in several scientific fields, such as biogeography, macroecology, physiology and genetics, to disentangle the intrinsic and extrinsic cues regulating internal clocks leading to the synchrony of gamete maturation and release.

Author Contributions

C.M. and V.L. conceived the study and designed the data collection strategy. C.M. curated and analysed the data and wrote the manuscript with contributions from S.A. V.L. secured funding. All authors participated in data collection and provided input for the final version of the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Site and colony scale spawning data including associated temperature data supporting the findings of this study can be downloaded from SEANOE at <https://www.seanoe.org/data/00947/105886/> (DOI: 10.17882/105886). The light data are provided in the [Supporting Information](#) (Appendix S6) of this article.

Code Availability

The code generating the analyses and figures is available on Dryad at <https://doi.org/10.5061/dryad.cz8w9gigs> (DOI: 10.5061/dryad.cz8w9gigs).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.